Heating & gas pressure

Warm or change state

1 Internal energy

Total kinetic + potential energy of all particles. Temperature is related to average kinetic energy.

$\Delta E = mc\Delta\theta$ when temperature changes; use kg, J/kg°C and a temperature difference.

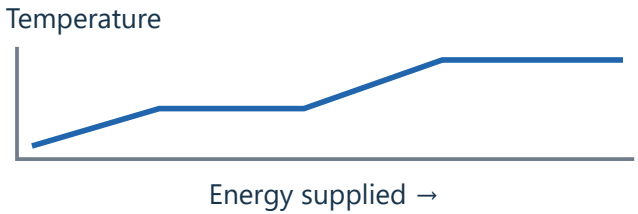
2 Latent heat

$E = mL$, L in J/kg. Change state without changing temperature.

Melting/boiling: potential energy rises. Freezing/condensing releases energy. c and L describe different changes.

Read energy changes

3 Heating graphs



Slopes: temperature rises. Plateaus: state changes while energy still enters.

Constant-power heating: longer time means more input energy. Cooling gives the reverse changes.

4 Combine stages

0.10 kg ice at 0°C → water at 20°C: mL + mcΔθ.

With L = 334,000 and c = 4,200, total = **41,800 J.**

At 100 W, ideal time = 418 s; losses add time.

Use collision models

5 Fixed-volume heating

Fixed mass: hotter → higher average kinetic energy → faster particles.

More frequent wall collisions and larger momentum changes → greater force per area → greater pressure.

6 Volume & compression

Fixed mass, constant T: **pV constant.** 100 kPa × 60 cm^3 = 150 kPa × 40 cm^3.

Rapid compression does work and may heat gas, so constant-T rule may not apply.